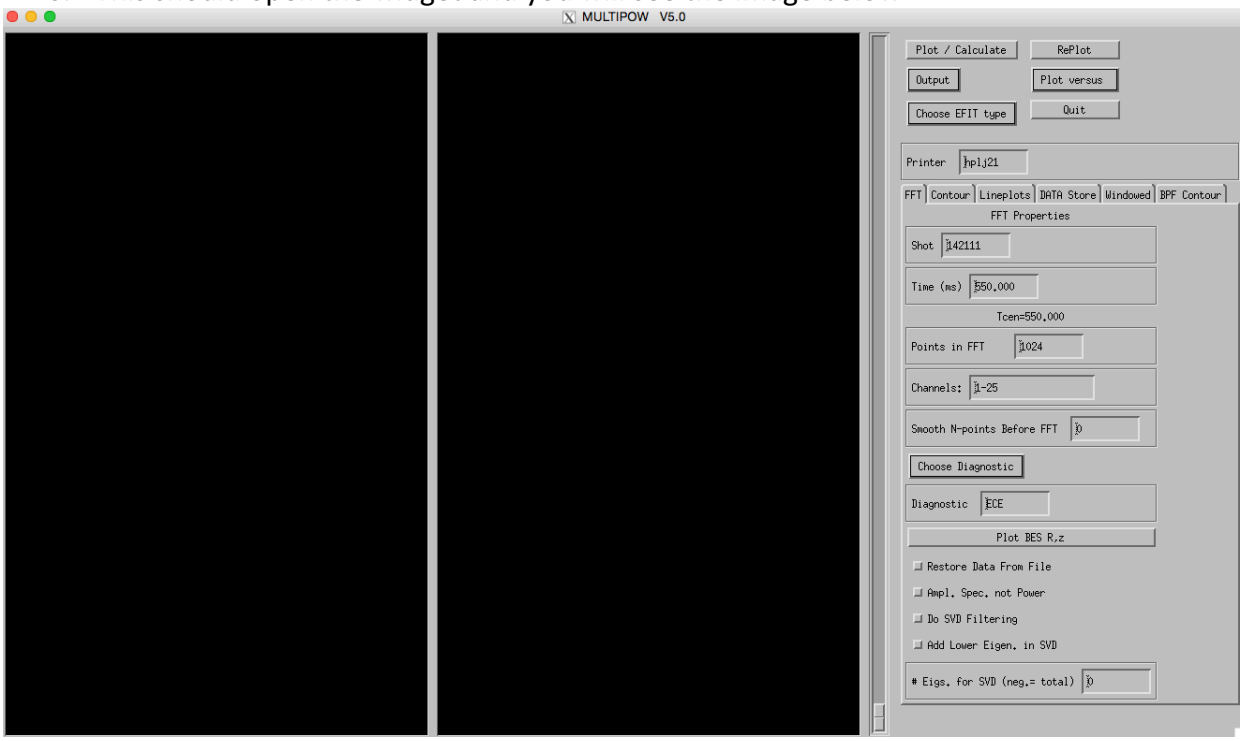
**MULTIPOW**
**2/6/19**
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**Background:** MULTIPOW is an IDL tool for analyzing the spatial structure of coherent fluctuations in ECE and BES data.

**Basic Introduction Using Fast ECE as an Example (Red = button, menu or input, Blue = Tab)**

1. Before using multipow, create a directory somewhere in your project area that you will run multipow from consistently. Multipow generates IDL .sav files and other output files and it is convenient to have a dedicated directory
2. cd to the run directory
3. module load ep
4. idl
5. IDL> multipow
6. This should open the widget and you will see the image below




7. **FFT Tab** – The FFT tab has most of the essential parameters for the analysis including the shot/time info, the FFT parameters and also which data are used.
    a. Shot = shot number
    b. Time (ms) = time at center of Fourier window
    c. Points in FFT = number of temporal points used in Fourier window
    d. Channels = Channels to use. ECE has 1-40 primary channels that the spatial information is stored for. BES has 64 possible channels. Channels can be specified as ##-##, ##-## to not display subsets if desired.

e. Smooth N-points Before FFT = **if the data are already stored locally,** setting this to a value of 3 or more will cause multipow to smooth the 2D (radius, time) array of data over the radial dimension by this many channels before doing Fourier analysis on the time dimension. This can really cut down on incoherent contributions to the signals but it does broaden modes.

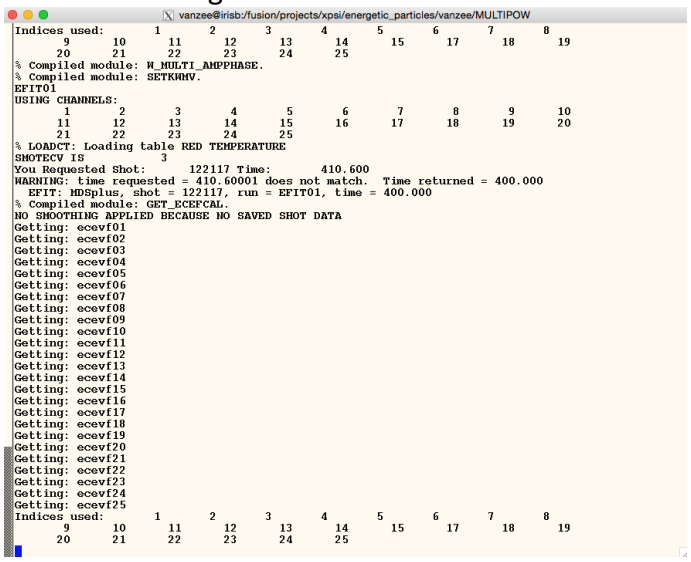
f. Choose Diagnostic = This is a pull down menu and allows you to select "ECE, ECES, BES". ECE is the 500kS/s ECE data, ECES is the 1kS/s ECE data and BES is the 2MS/s BES data. *note, if BES data are used, it is best to select Use dS/S not dS under the Lineplots tab.

g. Plot BES R,z = This is a push button that will display the BES R,z position data and channel number to make it easier for selecting the desired channels

h. Restore Data From File = rather than loading just the fourier interval for every desired channel, this will restore an idl .sav file saved in the local directory with the data saved over a larger time range. The .sav file must have been created beforehand using the "Data Store" tab discussed below.

i. Ampl. Spec. not Power = checking this will display amplitude vs. radius instead of power vs. radius.

8. After the above have been set, press the Plot / Calculate button
9. After the data have been loaded and Fourier analysis of each channel completed, the console should read something like below

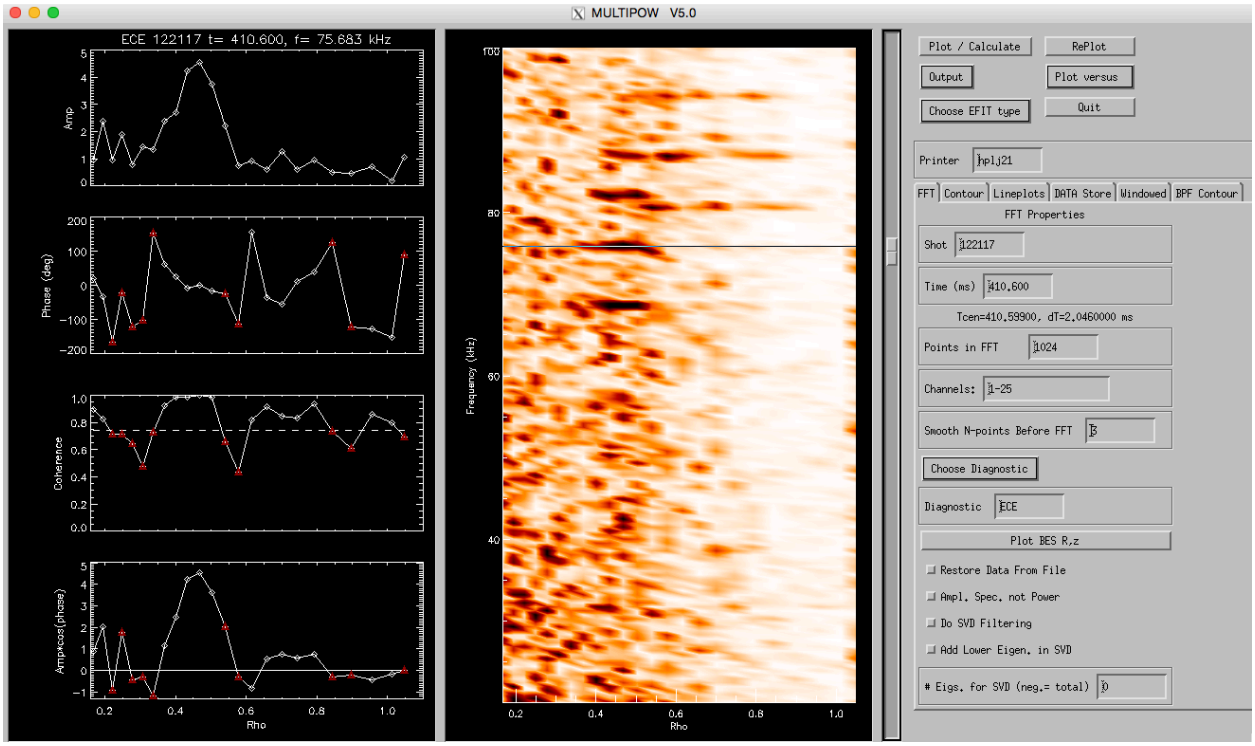



10. The middle window (Window 2) should now have a power (or amplitude) spectrum vs. Rho = normalized sqrt. Toroidal flux.

a. The radial coordinate can be changed between *Rho, Radius (cm)*, and *Channel* by selecting from the drop down menu labelled Plot versus

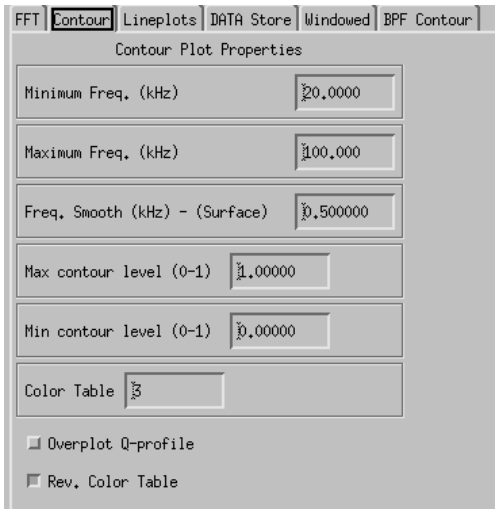
b. The mapping to position is accomplished using the MDSPlus stored EFIT tree which is selected under the Choose EFIT type pull-down menu. The closest EFIT to the center of the FFT time window is chosen.

11. By dragging the slider (between the center window and the input tabs) to the desired frequency, the spatial structure of the amplitude and phase data at that frequency and time can be visualized in the leftmost window (window 1). An example is shown below

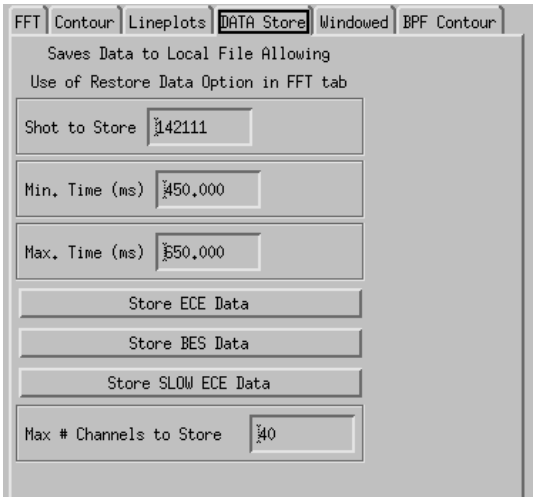



12. Window 1 contains, from top to bottom (plotted vs. desired radial coordinate):
    a. Fourier amplitude (for ECE, the units are eV)
    b. Phase relative to channel with peak amplitude.
    c. Coherence relative to peak channel (using a smoothing dF set on the Lineplots tab). Note, the 95% significance level is overplotted as a dashed line. All points with coherence below this level are indicated with a red point.
    d. Amplitude*cos(phase)

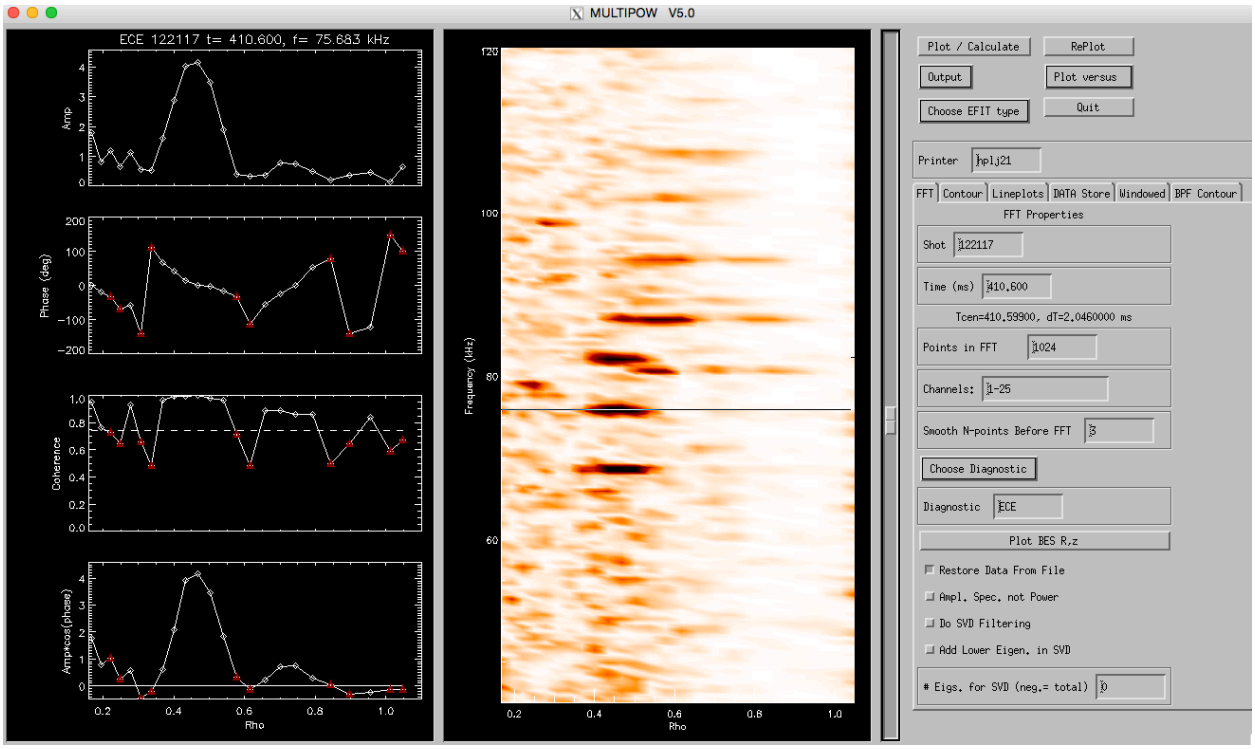
*13.* **Contour Tab**: The display properties of the contour in window 2 can be changed under the Contour tab. *If these are changed, press RePlot to have them take effect*
    a. Minimum Freq. (kHz) = minimum frequency plotted in contour
    b. Maximum Freq. (kHz) = maximum frequency plotted in contour
    c. Freq. Smooth (kHz) – (surface) = frequency smoothing interval before displaying of the contour plot
    d. Max and Min contour levels = can be used to saturate different parts of the color map
    e. Color Table = idl color table # (0-40)
    f. Overplot Q-profile = will overplot normalized q profile on contour
    g. Rev. Color Table = reverses color table when plotting contour



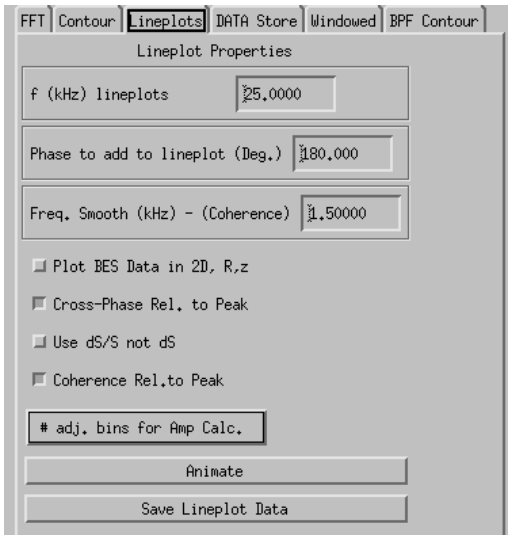

14. **Data Store Tab**: For faster analysis of a given shot and also to smooth data radially before Fourier analysis, the data can be stored locally using the Data Store tab
    a. Shot to Store = the shot which an .idl save file will be created for
    b. Min. Time (ms) = the beginning of the time window to save
    c. Max. Time (ms) = the end of the time window to save
    d. Max # Channels to Store = the data are saved starting with channel 1 and running to this number.
    e. Press the desired button corresponding to which data to be stored, i.e. "Store ECE Data", "Store BES Data" or "Store SLOW ECE Data".



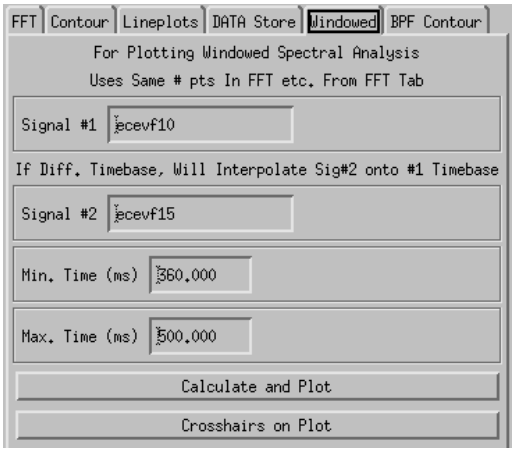

15. Once data are stored locally, the Restore Data From File button can be selected under the FFT tab. Also, smoothing can be done. The example below shows the results of changing the contour plotting window and smoothing over 3 points radially before carrying out the Fourier analysis. Noise is visibly reduced from that above.

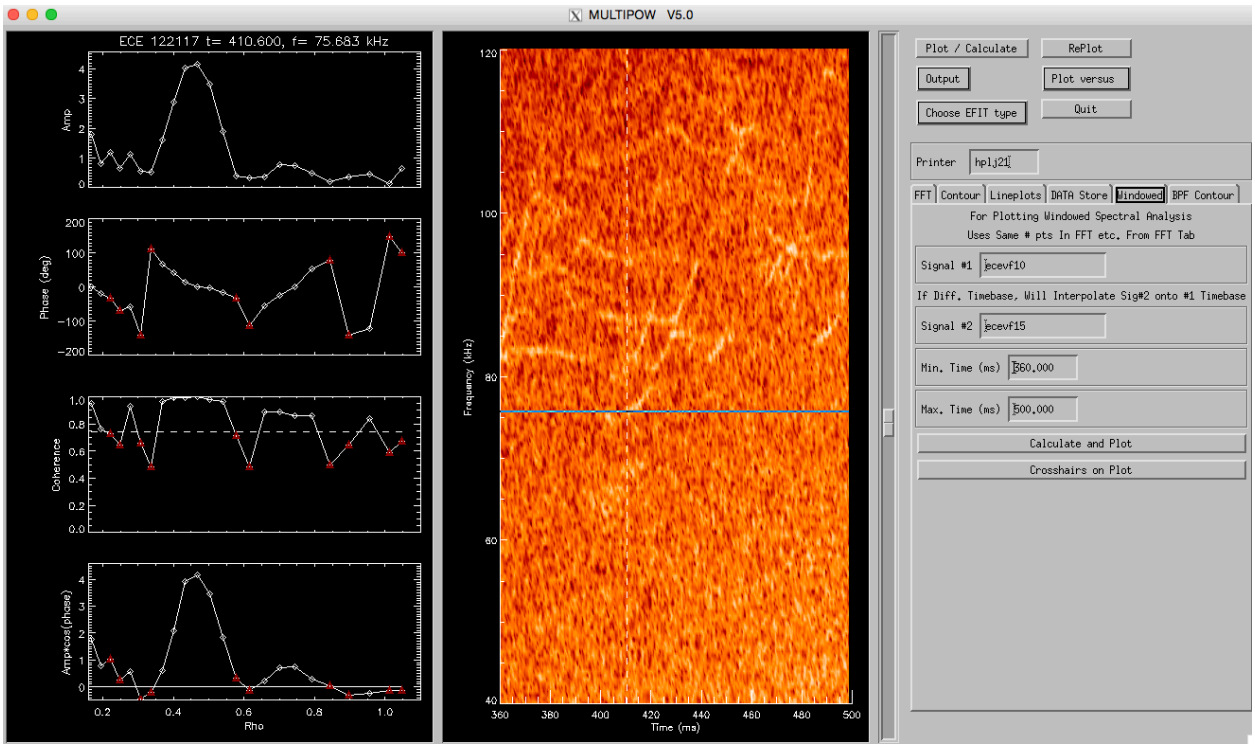



16. **Lineplots Tab:** Several inputs under the Lineplots tab can affect the lineplot data in Window 1.
    a. F (kHz) lineplots = the frequency to plot. Same effect as using the slider
    b. Phase to add to the lineplot (Deg.) = when plotting amp*cos(phase) in the lower panel, this phase shift will be added as: amp*cos(phase+deltaphase)
    c. Freq. Smooth (kHz) – (Coherence) = The frequency smoothing interval for the coherence calculation. If too few data points are chosen in the fourier interval, this has to be increased to encompass three or more frequency bins
    d. Plot BES Data in 2D, R,z = If 2D BES data are used, this will plot them in 2D rather than vs. a line
    e. Cross-Phase Rel. to Peak = selected by default
    f. Use dS/S not dS = when selected, this will divide all amplitudes by the $0^{th}$ frequency bin amplitude. For BES data, it normalizes to the Slow BES data.
    g. Coherence Rel. to Peak = selected by default
    h. # adj. bins for Amp. Calc. = this is an important selection that determines how many frequency bins are integrated over for the Fourier amplitude plotted in Window 1. If a large number of FFT points are chosen, a coherent mode can span multiple Fourier windows so by integrating multiple bins, a more accurate representation of the mode amplitude is obtained. The default is +/- 1 bin.
    i. Animate = this button will add a series of phase shifts in the Amp*cos(phase+deltaphase) plot, simulating the temporal behavior of the mode
    j. Save Lineplot Data = This will create an idl .sav file called **multi_ampphase.sav** to the local directory that contains all of the data plotted in windows 1 and 2.

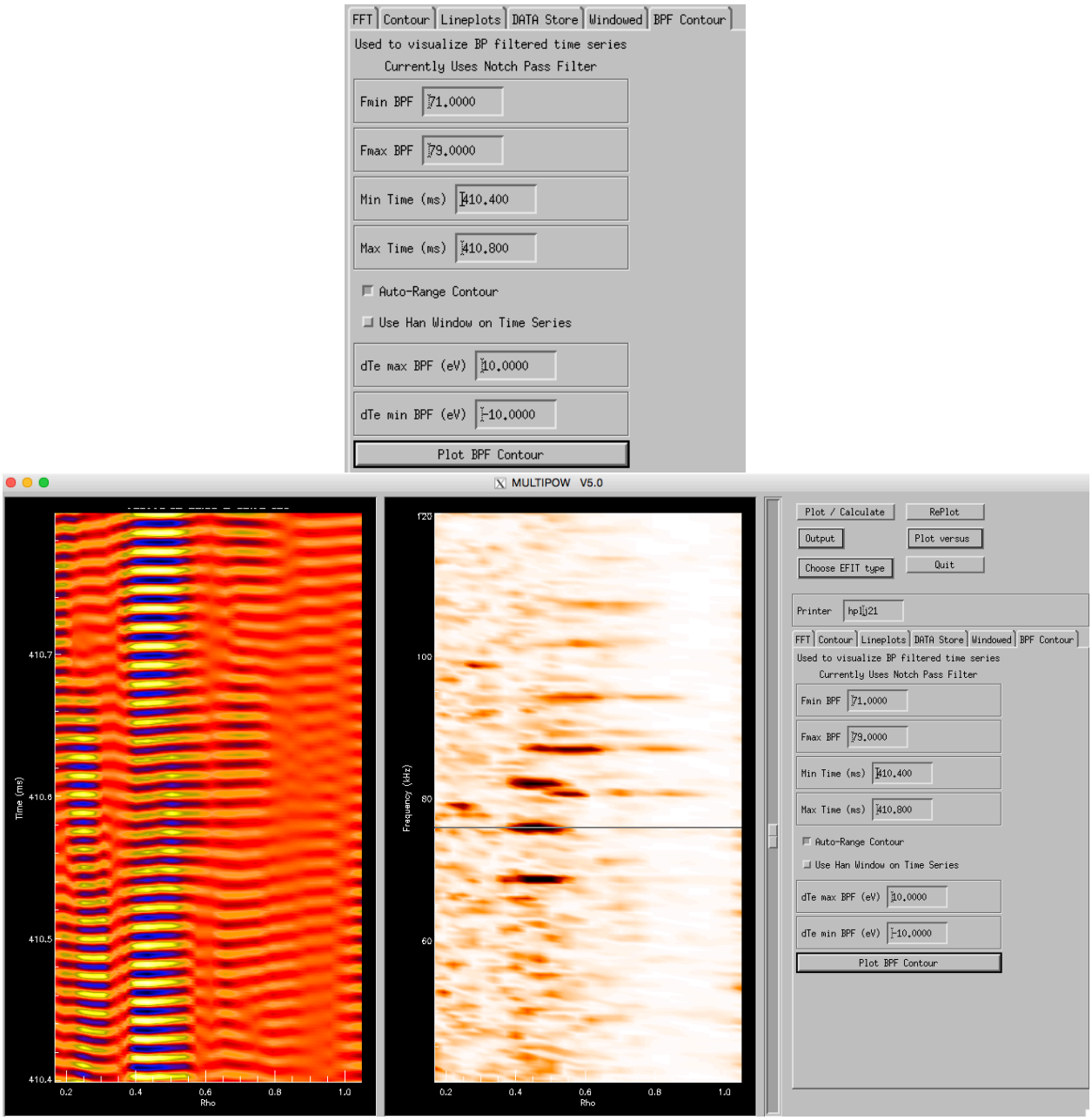



17. **Windowed Tab:** The Windowed tab allows the user to plot the crosspower of two signals over a specified time interval in Window 2. Note, the pointname for 500kS/s ECE data before shot 152721 was ECEVF## (##=01-40), after shot 152721, the pointname is ECEVS##. Note, the same number of points are used for the Fourier windows as set under the FFT tab.
    a. Signal #1 = First signal in crosspower
    b. Signal #2 = Second signal in crosspower. Note, if this pointname has a different timebase than the Signal #1, Signal #2 will be interpolated onto Signal #1's timebase.
    c. Min. Time (ms) = Beginning of time window for windowed crosspower spectrum
    d. Max. time (ms) = End of time window for windowed crosspower spectrum
    e. Calculate and Plot = generate the spectrogram and plot it in Window 2. The vertical dashed line is the center time of the Fourier window from FFT tab.
    f. Crosshairs on Plot = *First press Replot button, then Calculate and Plot*. Now, pressing Crosshairs on Plot will generate a set of crosshairs on the spectrogram. Left clicking will make it write the time and frequency to the console. Right clicking will exit crosshairs. This is useful for identifying the exact time desired for analysis of individual modes.

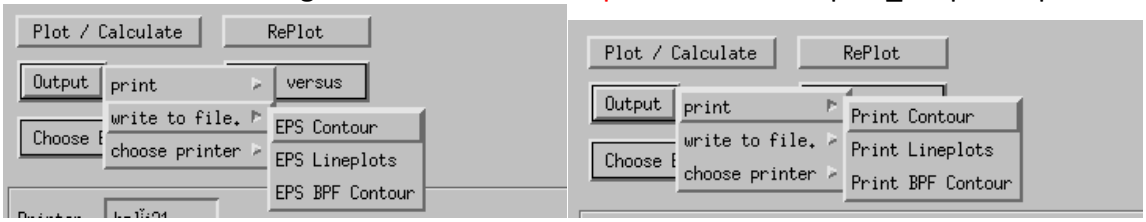





18. **BPF Contour Tab**: The BPF Contour tab allows the user to plot a contour of raw bandpass filtered data vs. radius. The filter is a simple notch filter. An example is shown below. The inputs under this tab are:
    a. Fmin BPF = lower frequency of the bandpass filter (kHz)
    b. Fmax BPF = upper frequency of the bandpass filter (kHz)
    c. Min Time (ms) = minimum time to plot of the filtered data
    d. Max Time (ms) = maximum time to plot of the filtered data
    e. Auto-Range contour = will automatically scale the colormap to the min/max of the plot
    f. Use Han Window on Time Series = Will use a Hanning window on the time series for filtering
    g. dTe max BPF (eV) = if Auto-Range Contour is not selected, this is the max in colormap
    h. dTe min BPF (eV) = if Auto-Range Contour is not selected, this is the min in colormap
    i. Plot BPF Contour = press this to filter and plot the data in Window 1.




19. The Output pull-down menu allows the user to print or generate .eps files of the various plots.
    a. Choosing write to file . EPS Contour writes multipow_contour.eps
    b. Choosing write to file . EPS Lineplots writes multipow_lineplots.eps



20. To exit, press Quit !

**Any comments, questions or suggestions, email Mike Van Zeeland (vanzeeland@fusion.gat.com)**